A Dataset Construction Details

We provide the complete pipeline for data collection, preprocessing, and annotation. This includes document source distribution, sampling criteria, and anonymization protocols. Each document is stripped of personally identifiable information and re-rendered into page-level PDFs for OCR benchmarking.

Document Sources and Sampling. Documents are collected from publicly available financial repositories, including the U.S. Securities and Exchange Commission (SEC) and open-source tax forms for non-profit organizations. We include five primary document types: *financial statements*, *required SEC filings*, *tax forms*, *securities transaction records*, and *financial legal documents*. To ensure annotation quality, each sampled filing must contain at least one type of critical fields.

Table 7: Financial Documents Type Distribution

Document Type	Data Source	Examples	Characteristics	Data size
Financial Statements	SEC EDGAR	10-K, 10-Q	Dense tables, mixture of modality and formats	44
SEC Required filings	SEC EDGAR	Form 4 , Form 8-K	Complex table structure	142
Tax forms	Candid	Form 990	Cross page tables and charts	11
Transaction Records	SEC EDGAR	Securities Transaction Records	Well structured table, numerically heavy	25
Financial Legal Documents	SEC EDGAR	M&A agreements, credit agreements, bond indentures	Long-form legal text with embedded financial tables, contract terms	637

Preprocessing and Rendering. All source documents are converted into canonical HTML using a rule-based extraction pipeline. Each HTML file is then paired with its corresponding page-level image deriving from intermediate PDFs, rendered at 300 dpi using Chromium headless mode to preserve layout fidelity. During rendering, embedded objects such as charts, tables, and formulas are retained as rasterized elements. For each document, metadata (document type, filing year) is recorded and stored in structured JSON format. To ensure compatibility with OCR models, we further normalize text alignment, table borders, and visual regions.

Annotation and Quality Control. Annotations are performed directly on HTML text, using the corresponding page image as visual reference for validation. Annotators highlight entities within HTML and embed specialized span tags encoding the five critical value types. An internal agreement table reports consistent reliability across both entity types (see Table 3 and appendix B).

Final Statistics and Accessibility. The final dataset contains 859 fully annotated financial documents, totaling 9,481 annotated entities.Each page is accompanied by its source metadata, HTML text, and annotated span-level markup. All files are distributed under a research-only license via a secure hosting repository.

B Data Annotation

FinCriticalED annotation procedure revolves around entity labeling for financial documents. The goal is to identify financially critical entities, focusing on numbers and time.

Annotation Guideline

```
Entity List
- Number
- Temporal
- Monetary Unit
- Reporting Entity
- Financial Concept

General Rules
- Identify all entities in the HTML file that belong to one of the five
  categories listed above.
- Use the rendered HTML as the primary source of truth; use the corresponding
  page image only for visual assistance when the layout or OCR text appears
  ambiguous.
- Highlight all valid entities in each task without omission.

Guidelines on entity type definition and label instructions
- Number
  - The numeric value should be financially critical, including percentages,
    monetary amounts, and share amounts.
  - Annotate only the number itself, or the number with attached signs, which
    may include decimal points, commas for thousands, and negative markers.
  - Do not include non-financially-related numbers, such as serial numbers or
    policy section numbers.
  - Examples: 1,000,000, 2,345, 0.37, 10, 1/3, -2.3, (10,234), 25.63%.
    Do not annotate examples such as Section 30(h).
- Temporal (Duration and Dates)
  - Only include specific dates and time periods. Do not include time
  frequency expressions such as monthly or weekly.
  - If the annotation target is a date range, annotate only the date itself
  and remove any context words between the two dates.
  - Examples: March 24, 2025, 1 month, and 2 years should be annotated.
  - For a span such as PERIODS JANUARY 2, 2025 THRU JANUARY 8, 2025, only
    annotate JANUARY 2, 2025 and JANUARY 8, 2025; exclude PERIODS and THRU.
- Monetary Unit
  - Monetary-related units should be included.
  - Examples: $, USD, thousands, millions, share.
- Reporting Entity
  - If the entity, such as a company name, contains the full name followed
  by the abbreviation, annotate them together.
  - Example: 1st Franklin Financial Corporation ("1FFC").
  - Additional examples: 1st FRANKLIN FINANCIAL CORPORATION, 1832 Asset
    Management L.P.
  - Include the whole entity, e.g., American Century Investment Management,
    Inc.
- Financial Concepts
  - Only label financial concepts if they appear inside a table or form.
  There is no need to label table titles or footnotes.
  - Label the whole line item if it contains a financial concept.
  - Example: Maximum Sales Charge (Load) Imposed on Purchases (as a
  percentage of offering price).
  - Label a term as Financial Concept if it belongs to one of the
  following categories:

    1. Revenue and Sales
       Concepts describing inflows from core business activities.
       Examples: Revenue, Net revenue, Gross revenue, Sales, Net sales,
       Service revenue, Total Revenue.
    2. Income, Profit, and Earnings
       Concepts describing profitability or earnings outcomes.
       Examples: Income, Net income, Operating income, Gross profit,
       Profit, Earnings, Earnings per share (EPS).
    3. Costs, Expenses, and Losses
       Concepts describing outflows, reductions, or negative performance.
       Examples: Cost, Cost of revenue, Cost of sales, Operating expenses,
       Expense, Loss, XXX Fee, XXX Expenses.
    4. Taxes
       Concepts related to taxation and tax-related outcomes.
       Examples: Income tax, Tax expense, Deferred tax, Tax liability,
       Effective tax rate.
    5. Margins and Ratios
       Common, high-level ratios directly tied to revenue or profit.
       Examples: Gross margin, Operating margin, Profit margin,
       Net Income/Loss.
    6. Financial Obligations and Commitments
       Concepts describing required or expected payments.
       Examples: Lease obligation, Purchase obligation, Debt,
       Interest expense, Interest income.
  - Explicit Exclusions (Do NOT label)
    1. Operational or Functional Activities:
       Research and Development, Marketing, Sales and distribution,
       General and administrative.
    2. Accounting or Reporting Metadata:
       Accounting policy, Notes to the financial statements,
       Segment information, Management discussion and analysis (MD&A).

    3. Broad Business or Strategy Terms:
       Growth strategy, Market expansion, Customer acquisition,
```

```
      Product roadmap.
   4. Non-Financial Metrics or Qualitative Descriptions:
      Headcount, Employee engagement, ESG initiatives.
```

B.1 Annotator Demography

FinCriticalED was annotated by a four-person team with complementary expertise in finance, economics, auditing, and computer technology. The team combines professional experience in financial analysis and FinTech with graduate-level training in business analytics, financial mathematics, computer science, and computer technology.

One annotator is a principal analyst at a major U.S. financial institution with training in business analytics, statistics, and economics, as well as experience in LLMs, financial data analysis, and multilingual reasoning. Another annotator has training in financial mathematics and computer science, along with more than seven years of experience in strategic finance and consulting in the FinTech industry. The remaining two annotators are graduate students in computer technology with backgrounds in auditing, financial analysis, data processing, annotation workflows, and LLM evaluation and adaptation for domain-specific tasks.

Overall, the team reflects a balance between financial domain knowledge and technical expertise, supporting accurate, consistent, and contextually grounded annotation across the dataset.

B.2 Annotation Process

The annotation workflow of FinCriticalED follows a structured, multi-stage process designed to ensure factual precision and consistency across annotators. Each annotator works within a controlled web-based interface displaying paired HTML text and its corresponding rendered page image. This dual-view setup lets annotators reference the visual layout when verifying line breaks, superscripts, or numerical formatting inconsistencies.

The annotation process begins with a **pre-screening phase**, where annotators review extracted HTML text for structural integrity and identify potential OCR or layout errors. Once validated, they proceed to the **entity marking phase**, where relevant critical fields are highlighted and enclosed within specialized span tags (e.g., `<Number>...</Number>`).

20% of the data is annotated independently by at least two annotators to measure inter-annotator agreement. Upon completion, annotations are exported into JSON format, preserving the hierarchical document structure and linking each annotated entity to its source metadata. This structure enables downstream comparison against model-generated predictions for entity-level evaluation.

B.3 Validation Guideline

To ensure reliability and reproducibility, FinCriticalED employs a multi-layered validation protocol combining automated checks, cross-annotator comparison, and expert review.

First, an **automated integrity validation** step scans all annotated files to detect malformed or nested span tags, missing entity attributes, or misaligned indices within the exported JSON schema.

Second, an **inter-annotator consistency check** evaluates pairwise agreement using token-level and entity-level overlap metrics.

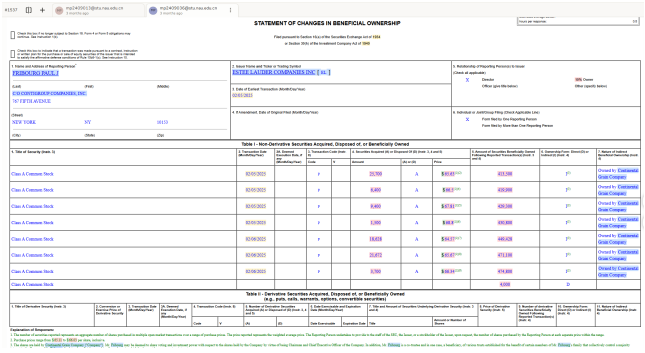


Figure 4: Annotation interface used in FinCriticalED. Annotators highlight entities directly within HTML while referencing rendered page images for layout validation.

Entities with disagreement scores below a fixed confidence threshold are automatically queued for adjudication. Discrepancies are resolved through a consensus meeting facilitated by a lead annotator, who applies majority agreement rules and reviews edge cases (e.g., ambiguous table headers or multi-line numeric ranges).

Third, a **domain validation review** is conducted after every 100 annotated samples. In this phase, a senior financial expert randomly samples 5% of the annotations to assess factual correctness. Feedback from this phase is incorporated into a continuously refined annotation guideline that codifies new patterns or exceptions encountered during labeling.

Finally, the validated annotations are version-controlled and re-exported to ensure traceability across dataset releases. The combination of automated validation, human consensus, and domain-level oversight establishes a high-confidence ground-truth benchmark for evaluating factual accuracy and numeric fidelity.

B.4 Annotation Agreement Metric

To assess the reliability and consistency of human annotation in FinCriticalED, we employ both **Cohen's Kappa** [38] for pairwise annotator agreement and **Fleiss' Kappa** [40] for overall multi-annotator reliability. These chance-corrected metrics provide a robust measure of agreement that accounts for the likelihood of random coincidence in categorical labeling.

Cohen's Kappa (κ) quantifies the level of agreement between two annotators and is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{1}$$

where p_o represents the observed agreement (the proportion of items on which both annotators agree) and p_e denotes the expected agreement based on random chance. A value of $\kappa = 1$ indicates perfect agreement, while $\kappa = 0$ reflects agreement equivalent to chance.

For more than two annotators, we adopt Fleiss' Kappa, which generalizes the formulation of inter-rater agreement to multiple raters. It is computed as:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}, \tag{2}$$

where $\bar{P}$ denotes the mean proportion of observed agreement across all annotation units, and $\bar{P}_e$ indicates the expected probability of agreement under a random labeling assumption. Similar to Cohen's Kappa, higher values signify stronger consensus among annotators.

In FinCriticalED, we calculate Cohen's Kappa for each annotator pair to evaluate pairwise consistency and Fleiss' Kappa for the full group to measure overall reliability across five entity categories. Statistics in table 3 confirm a high level of agreement among the annotators, underscoring the robustness of the dataset's annotation process.

C Deterministic-Rule-Guided LLM-As-Judge

Pipeline Overview. The evaluation proceeds in three stages:

(1) *Entity Extraction*: *BeautifulSoup* parses the gold HTML to enumerate all annotated entities by type, each paired with a ±40-character context hint;
(2) *LLM Judgment*: GPT-4o (`temperature=0`) receives the entity list, stripped gold text, and model prediction, and returns a JSON object indicating per entity whether it was *found* and *exactly matched*;
(3) *Deterministic Override*: any entity passing a normalized exact-match check is unconditionally set to correct = `true`, regardless of the LLM decision.

LLM-as-Judge prompt

```
JUDGE_PROMPT_TEMPLATE = """
#Instruction: You are an expert OCR results inspector for financial
documents. You will be given:
1. A pre-extracted list of financially critical entities from the ground
truth HTML. Each entity has: - "type": one of "Number", "Temporal",
"Monetary Unit", "Reporting Entity", "Financial Concepts"
   - "value": the exact text of the entity
   - "context_hint": a short fragment of surrounding text that helps locate
   it in the document
2. Ground truth HTML (entity tags already stripped, plain structure only)
— for reference context.
3. Model-generated HTML that was produced from the same image.
Your task: for each entity in the pre-extracted list, determine whether
the model-generated HTML contains that entity.
# Two-level judgment per entity
For each entity output TWO boolean fields:
- "found_in_prediction": true if the entity's value appears anywhere
in the model HTML in a recognizable form — even if slightly corrupted
(e.g., missing  comma, merged spaces, truncated).
  Set to false only if the entity is completely absent from
  the prediction.
- "correct": true ONLY if the entity satisfies the strict
matching rules below.
  (correct=true implies found_in_prediction=true;
  found_in_prediction=true does NOT imply correct=true)
# PRIORITY RULE: exact match → always correct
If the matched_text (whitespace-normalized, case-insensitive)
is identical to value, set correct=true.
The context_hint is only used to DISAMBIGUATE when the same
value string appears in multiple different
locations in the prediction. It is NOT a veto: a different surrounding
context does NOT make an otherwise exact match incorrect.
# Context-hint matching (apply loosely, only for disambiguation)
The context_hint is extracted from annotated HTML and may have minor
formatting differences vs the prediction:
- Spaces may appear around currency symbols in the hint (e.g.,
"$ 66.59") but not in the prediction ("$66.59").
- Commas around numbers may differ in spacing (e.g., "per share,
inclusive" vs "per share,inclusive").
- Apply whitespace-normalization and ignore spacing around $,
%, (, ) when comparing context.
- The entity value itself (e.g., "66.59") must still appear in or
very near the matched context region.

- IMPORTANT: Do NOT reject a match just because the context_hint
has different spacing than the prediction.
  If the surrounding words match (e.g., "Purchase prices range
  from", "per share, inclusive"), that is
  sufficient context confirmation — minor punctuation/spacing
  differences are expected and acceptable.
# Strict matching rules (apply when setting "correct")
- For "Number", "Temporal", "Monetary Unit":
  * correct=true if the exact numeric/date string (whitespace-normalized
  , case-insensitive) appears in
    the model HTML. If it appears multiple times, use context_hint
    to pick the right occurrence.
  * Do NOT mark correct if the string is corrupted, truncated, or
  merged with adjacent characters.
    Examples of NOT correct: "ERIBOURGPAUL" for "FRIBOURG PAUL J"
    ; "25700" for "25,700"; "0.5" for "0.50".
  * Mark found_in_prediction=true if a close but not exact version
  appears (e.g., "25700" for "25,700").
- For "Reporting Entity", "Financial Concepts":
  * correct=true if the full string appears (whitespace-normalized
  , case-insensitive) in the model HTML.
    If it appears multiple times, use context_hint to pick the
    right occurrence.
  * Minor run-together spaces within a single token (e.g.,
  "UNITEDSTATES" for "UNITED STATES") may be
    marked correct if still fully recognizable — record actual
    text in matched_text.
  * Missing words, swapped words, or substituted characters →
  correct=false, but may be found_in_prediction=true.
# Step 1: Match entities
For each entity in the pre-extracted list, output a diagnostic record
with:
  - "type", "value", "context_hint" (copied from input)
  - "found_in_prediction": true/false (recognizable attempt exists
  in prediction)
  - "correct": true/false (exact match per strict rules above; exact
  match always overrides context check)
  - "matched_text": the actual text found in the model HTML (empty
  string "" if not found at all)
# Step 2: Compute counts
The total_entities counts come directly from the pre-extracted list
— do NOT re-count from the ground truth HTML.
Compute:
  - found_entities = number of diagnostics where found_in_prediction
  = true
  - correct_entities = number of diagnostics where correct
  = true
  - found_entities_with_X_type and correct_entities_with_X_type
  for each of the 5 types
  - entity_accuracy = correct_entities / found_entities * 100,
  rounded to 2 decimal places (0.0 if found_entities = 0)
# Step 3: Output
Output exactly one valid JSON object:
{
  "total_entities": <int>,
  "total_entities_with_Number_type": <int>,
  "total_entities_with_Temporal_type": <int>,
  "total_entities_with_Monetary_Unit_type": <int>,
  "total_entities_with_Reporting_Entity_type": <int>,
  "total_entities_with_Financial_Concepts_type": <int>,
  "found_entities": <int>,
  "found_entities_with_Number_type": <int>,
  "found_entities_with_Temporal_type": <int>,
  "found_entities_with_Monetary_Unit_type": <int>,
  "found_entities_with_Reporting_Entity_type": <int>,
  "found_entities_with_Financial_Concepts_type": <int>,
  "correct_entities": <int>,
  "correct_entities_with_Number_type": <int>,
  "correct_entities_with_Temporal_type": <int>,
  "correct_entities_with_Monetary_Unit_type": <int>,
  "correct_entities_with_Reporting_Entity_type": <int>,
  "correct_entities_with_Financial_Concepts_type": <int>,
  "entity_accuracy": <float 0-100>,
  "overall_explanation": "<1-2 sentences summarizing model quality>",
  "entity_diagnostics": [
    {"type": "...", "value": "...", "context_hint": "...",
     "found_in_prediction": true, "correct": false, "matched_text": "25700"},
    ...
  ]
}
Rules:
* Output only valid JSON (no extra text).
* All count fields must be integers, not strings.
* entity_accuracy must be a float (e.g. 85.71), NOT a string like "85.71%".
* entity_accuracy =
correct_entities / found_entities * 100 (NOT correct / total).
```

```
* total_entities counts MUST match the counts from the pre-extracted list.
* REMINDER: if matched_text equals value (normalized), correct MUST be true —
do not set correct=false just because the surrounding context in the
prediction differs from context_hint.
"""
```

LLM-as-Judge output sample


```
{'total_entities': 65,
 'total_entities_with_Number_type': 37,
 'total_entities_with_Temporal_type': 10,
 'total_entities_with_Monetary_Unit_type': 7,
 'total_entities_with_Reporting_Entity_type': 11,
 'total_entities_with_Financial_Concepts_type': 0,
 'found_entities': 64,
 'found_entities_with_Number_type': 36,
 'found_entities_with_Temporal_type': 10,
 'found_entities_with_Monetary_Unit_type': 7,
 'found_entities_with_Reporting_Entity_type': 11,
 'found_entities_with_Financial_Concepts_type': 0,
 'correct_entities': 64,
 'correct_entities_with_Number_type': 36,
 'correct_entities_with_Temporal_type': 10,
 'correct_entities_with_Monetary_Unit_type': 7,
 'correct_entities_with_Reporting_Entity_type': 11,
 'correct_entities_with_Financial_Concepts_type': 0,
 'entity_accuracy': 98.36,
 'overall_explanation': 'The model successfully identified and correctly
 matched nearly all entities, with only one minor discrepancy in a number
 format.',
 'entity_diagnostics': [{'type': 'Number',
   'value': '10%',
   'context_hint': 'X Director 10% Owner Officer (give title below) Other',
   'found_in_prediction': True,
   'correct': True,
...
 'not_found_entities_with_Number_type': 1,
 'not_found_entities_with_Temporal_type': 0,
 'not_found_entities_with_Monetary_Unit_type': 0,
 'not_found_entities_with_Reporting_Entity_type': 0,
 'not_found_entities_with_Financial_Concepts_type': 0}
```


Design Details. Context Window. Many financial documents repeat identical values across different line items (e.g., the number "1,200" may appear in both revenue and headcount rows of the same table). To prevent the judge from conflating these, each entity is accompanied by the 40 characters immediately preceding and following it in the gold text. For instance, the entity `1,200` might be disambiguated by the hint "`...total revenue (in millions) 1,200 for the fiscal year...`", directing the judge to verify that specific occurrence rather than any surface match.

Normalization and Deterministic Override. Before applying the override, both the predicted and gold entity values are normalized: lowercased, whitespace-collapsed, and stripped of punctuation (commas, currency symbols, trailing periods). If the normalized strings are identical, the entity is forced correct. This guards against LLM false negatives on unambiguous cases.

Found vs. Correct. An entity is *found* if the judge locates it (or a semantically equivalent expression) anywhere in the prediction; it is *correct* if the reproduced value is additionally judged to be factually exact. An entity may be found but incorrect (e.g., a number located in the wrong context), or neither found nor correct (entirely absent from the prediction).

Human Alignment Validation. To validate the automated judge, 15 outputs from Paddleocrv5, gpt-4o, and gpt-5 each (45 total) were independently evaluated by a human expert using the same found/correct schema. We adopt *Document-Level Strict Agreement*: judge and human must agree on every entity within a document to count as aligned; a single divergence constitutes disagreement. GPT-4o achieved **95%** agreement after iterative prompt and evaluation workflow refinement. Figure 5 shows a representative alignment example:

Case: Overall FFA = 90%. On Figure 5, both human experts and the LLM-as-Judge classify the output as factually unreliable. The LLM-as-Judge hallucinates the row name "Certificate rate" to "Interest rate," as it associate with the "Interest paid" row name below. The model also misread the monetary amount. *LLM output:*


```
{
    "total_entities": 23,
    ...
    "found_entities_with_Number_type": 3,
    ...
    "found_entities_with_Financial_Concepts_type": 10,
    ...
    "correct_entities_with_Number_type": 2,
    ...
    "correct_entities_with_Financial_Concepts_type": 9,
    "entity_accuracy": 90.0,
    "overall_explanation": "The model accurately identified most entities,
    especially reporting entities and financial concepts, but struggled
    with exact numeric matches."
    }
```


Human expert explanation:

```
Total entities: 23
Numeric entities: 3
...
Financial Concepts: 10
...
Wrong numeric entities: 1
...
Wrong reporting entities: 1
Overall:  The model performs well overall, except errors due to halluciantion
on "Interest rate" and wrong numerical entity of "21,829,000
(should be "21,429,000").
```

These observations correspond precisely to the errors and overall judgment from GPT-4o output.

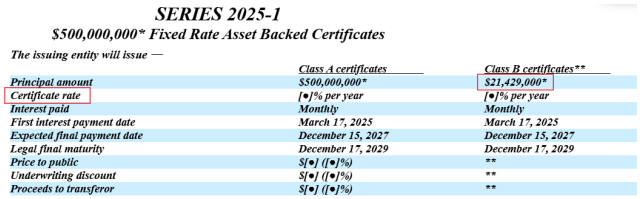
D Model Failure Analysis

This section analyzes model failures beyond overall FFA scores. We first describe the error-analysis setup, then examine two notable aggregate patterns: elevated missing-output behavior in some proprietary models and GLM-OCR's strong performance on financial concepts. We finally present representative error cases illustrating how different failure modes appear in practice.

D.1 Error Analysis Setup

We perform error analysis by comparing annotated ground-truth content with model-predicted OCR outputs after both are rendered in HTML. Specifically, the ground-truth document and the corresponding model output are loaded into Label Studio, where annotators inspect them side by side and identify financially meaningful discrepancies. During review, they assess whether a model preserves both local transcription and the financial meaning of each critical field under its rendered context, rather than only surface-level textual overlap.

We organize reviewed examples along two dimensions: **document modality** and **document complexity**. We first categorize documents into three modality types:



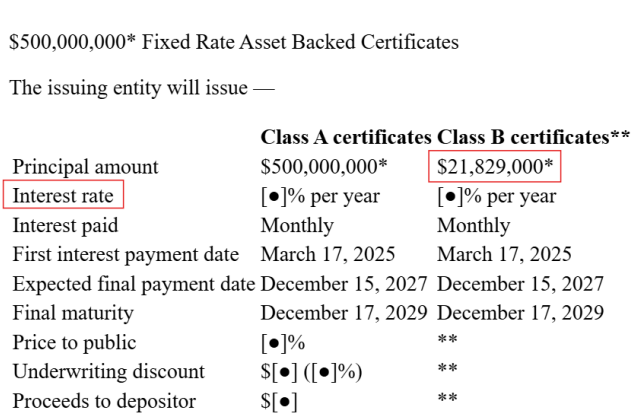
SERIES 2025-1

$500,000,000 Fixed Rate Asset Backed Certificates*

The issuing entity will issue —

	Class A certificates	*Class B certificates***
Principal amount	*$500,000,000**	*$21,429,000**
Certificate rate	*[•]% per year*	*[•]% per year*
Interest paid	*Monthly*	*Monthly*
First interest payment date	*March 17, 2025*	*March 17, 2025*
Expected final payment date	*December 15, 2027*	*December 15, 2027*
Legal final maturity	*December 17, 2029*	*December 17, 2029*
Price to public	*$[•] ([•]%)*	****
Underwriting discount	*$[•] ([•]%)*	****
Proceeds to transferor	*$[•] ([•]%)*	****

(a) Part of financial statement investment plan explained



SERIES 2025-1

$500,000,000* Fixed Rate Asset Backed Certificates

The issuing entity will issue —

	Class A certificates	**Class B certificates****
Principal amount	$500,000,000*	$21,829,000*
Interest rate	[•]% per year	[•]% per year
Interest paid	Monthly	Monthly
First interest payment date	March 17, 2025	March 17, 2025
Expected final payment date	December 15, 2027	December 15, 2027
Final maturity	December 17, 2029	December 17, 2029
Price to public	[•]%	**
Underwriting discount	$[•] ([•]%)	**
Proceeds to depositor	$[•]	**

(b) Corresponding LLM output in HTML

Figure 5: Human alignment with evaluation results on one of the GPT-4o outputs. Highlighted areas indicate where OCR results on critical fields are incorrect.

- **Table-only**: primarily tabular content, including table of contents pages or documents containing only tables, with no surrounding text beyond titles or headers.
- **Text-only**: contain only text without tables, including one or more paragraphs, with or without section headings.
- **Mixed modality**: contain both tables and text, where the text extends beyond titles, headers, or annotations.

Within each modality, we further divide examples into low, medium, and high complexity using modality-specific criteria:

- **Table-only**: low = simple tables such as table of contents pages; medium = clearly structured financial tables organized in a straightforward row-by-row format; high = complex or irregular financial tables, such as tax forms, where information cannot be easily interpreted line by line.
- **Text-only**: low = plain text without section headings or numeric financial indicators such as numbers, dates, or monetary values; medium = text with either section headings or numeric indicators such as numbers, dates, or monetary values; high = text with both section headings and numeric financial indicators.
- **Mixed modality**: low = documents containing tables with simple accompanying text without section headings or additional numeric indicators; medium = documents containing tables and text with either section headings or numeric indicators; high = documents containing tables and text with both section headings and numeric indicators.

Using this setup, we identify representative error cases and map them to the broader qualitative taxonomy introduced in Table 8.

D.2 Missing-Output Effects in Proprietary Models

GPT-4o and GPT-5 exhibit unusually low FFA partly because they have much higher missing-output rates than most other models in our evaluation. Their overall missing rates are 23.12% and 13.41%, respectively, compared with 1.94% for Qwen3-VL-8B-Instruct, 1.76% for Llama-4-Maverick, and 1.73% for Claude-Sonnet-4.6. The gap is especially large for numeric fields, where the missing rates rise to 31.35% for GPT-4o and 24.0% for GPT-5. In manual inspection, many failures appear as incomplete or prematurely cut-off HTML rather than straightforward recognition errors. Because this pattern persists even after substantially increasing the output budget, we do not attribute it mainly to length limits. Instead, it may reflect more conservative generation behavior on long, structured, and potentially sensitive financial pages, which is consistent with OpenAI's description of GPT-5 safe-completion and stronger caution under ambiguous-risk settings, while also suggesting a model tendency to abstain under difficult document conditions [10, 22].

D.3 Why Does GLM-OCR Reach 100% FC-FFA?

GLM-OCR on financial concepts. GLM-OCR achieves **100% FC-FFA** in our evaluation, indicating exceptional extraction of financial-concept spans such as line items and table labels. Because financial concepts are typically expressed as plain-text semantic units rather than symbol-sensitive values, performance on this field type depends heavily on robust text recognition and structure-aware parsing. This result is likely related to GLM-OCR's OCR-specialized design: a compact encoder–decoder architecture with a dedicated visual encoder, a GLM decoder, and a two-stage pipeline that first performs layout analysis and then applies region-level recognition [4]. This design is particularly well aligned with financial-concept extraction, where accurate isolation of row labels and preservation of local document structure are critical. More broadly, the result suggests that OCR-specialized document models can hold a clear advantage over general-purpose multimodal models on plain-text financial semantics, especially when success depends more on structural fidelity than symbolic precision.

D.4 Representative Model Error Cases

In Table 8, we presents the error taxonomy and the representative models and model failure examples:

E Limitations

While FinCriticalED advances fact-level evaluation for financial OCR, several limitations remain.

First, FinCriticalED mainly covers U.S. financial documents with relatively standardized visual conventions; broader international, multilingual, and handwritten settings may introduce challenges not captured here.

Second, our annotation pipeline uses rendered HTML as the structural reference. Although stable and reproducible, it may differ from native PDF or scanned-document artifacts in real OCR deployments.

Table 8: Qualitative taxonomy of observed Model OCR failure patterns. OCR-specialized systems tend to show fixed and repetitive recognition errors, while general multimodal LLMs exhibit more variable and generative failure modes.

Error Type	Description	Common Models	Example
Header misalignment	Correct value is transcribed but attached to the wrong column or local header context.	Llama-family	"6. Ownership Form: Direct (D) or Indirect (I) (Instr. 4)" value aligned to "7. Nature of Indirect Beneficial Ownership (Instr. 4)".
Heading recognition error	The model fails to correctly recognize section titles or subtitles, either by omitting them or transcribing them incorrectly, leading to incorrect document structure or emphasis.	All with different levels of severity	gemma-3n-E4B-it failed to recognize section subtitle "Credit card rates may decline without a corresponding change in the amounts needed to pay the certificates, which could result in a delay or reduction in payments of your certificates."
Hallucinated continuation	Similar sentence prefix causes the model to copy the continuation of a previously seen sentence instead of the correct local text.	Gemma-3n-E4B-it, GPT-4o	Document started to repeating the sentence "the undergaining and account management standards of card issuers and could restrict the ability of AEN's of its affiliates to take" on the last paragraph.
Fixed lexical corruption	The same word or token is repeatedly mistranscribed across pages.	DeepSeek-OCR, MinerU2.5	"transferor" was changed into "transfessor", "transforor", "transferror", or "transfeder" etc. across different documents.
Format-sensitive spacing	OCR errors consistently occur around dashes, punctuation, or spacing-sensitive formats.	DeepSeek-OCR, MinerU2.5	"Certain Legal Aspects of the Receivables — Consumer Financial Products Regulation" was changed into "Certain Legal Aspects of the Receivables - Consumer Financial Products Regulation"
Sentence-local corruption	An OCR error confined to a single sentence, where the model omits, inserts, rewrites, or distorts content within that sentence without affecting other parts of the document.	Llama-family	Document missed "and the collateral securing the certificates"
Unstable free-form corruption	Errors vary substantially across samples, even under similar layouts or textual contexts.	Gemma-3n-E4B-it, GPT-4.o, GPT-5	GPT-5 missed the whole paragraph that even including financial facts.

Third, FinCriticalED evaluates factual correctness at the entity level, but does not yet capture cross-page linking, hierarchical financial relationships, or long-context numerical grounding across document collections.

These limitations suggest future work in multilingual, hybrid human–machine evaluation for edge cases, and richer benchmarks combining OCR, layout understanding, and financial reasoning.

F Potential Risks and Misuse

FinCriticalED is intended to advance high-precision financial OCR research, but several risks should be noted.

First, although the benchmark uses publicly available financial documents, stronger OCR may enable more effective extraction of sensitive information from documents not intended for automated analysis. Responsible use requires compliance with privacy, regulatory, and data-handling requirements.

Second, the LLM-as-Judge framework is designed only for evaluation and should not be treated as an authoritative tool for auditing, compliance, or legally binding document verification.

Third, like any dataset, FinCriticalED may reflect geographic, regulatory, and formatting biases in its source documents. Models trained or tuned only on this dataset may overfit to these conventions and generalize poorly elsewhere.

Finally, highly accurate OCR may enable large-scale financial data extraction for questionable uses, including unauthorized scraping, adversarial market strategies, or amplification of misleading financial narratives. Researchers and practitioners should consider both beneficial and harmful downstream impacts.

G Ethical Considerations and Licensing

All documents come from publicly available financial filings released under open-access or research licenses. No proprietary or confidential information is included. The dataset is intended for research on document understanding and factual accuracy; commercial use requires separate compliance review. The benchmark complies with ACM data ethics policies.

Supplementary References

[38] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement* 20, 1 (1960), 37–46.
[4] Shuaiqi Duan, Yadong Xue, Weihan Wang, Zhe Su, Huan Liu, Sheng Yang, Guobing Gan, Guo Wang, Zihan Wang, Shengdong Yan, Dexin Jin, Yuxuan Zhang, Guohong Wen, Yanfeng Wang, Yutao Zhang, Xiaohan Zhang, Wenyi Hong, Yukuo Cen, Da Yin, Bin Chen, Wenmeng Yu, Xiaotao Gu, and Jie Tang. 2026. GLM-OCR Technical Report. arXiv:2603.10910 [cs.CL] https://arxiv.org/abs/2603.10910
[40] Joseph L. Fleiss. 1971. Measuring nominal scale agreement among many raters. *Psychological Bulletin* 76, 5 (1971), 378–382.
[10] Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, et al. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276* (2024).
[22] OpenAI. 2025. Gpt-5 system card.